

ECON 6018

TUTORIAL 7

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Optimization

• Step 1: FOC = 0

• Step 2: SOC

- $f''(x) > 0$ min

$f''(x) < 0$ max

Example:

$$f(x) = 4x^2 + 7$$

Practices : Find Min or Max

1. $g(x) = 3(x-1)^4$

2. $h(x) = \sqrt{2+x^4}$, $x \in [-1, 1]$

3. $F(x) = -\frac{3}{1+x^2}$

4. $G(x) = 3 - \sqrt{2-x}$

5. $H(x) = 50 - e^{-2x^2}$

Answers

1. $g(x) = 3(x-1)^4$

$$g'(x) = 12(x-1)^3 \quad x=1$$

$$g''(x) = 36(x-1)^2 \quad \therefore \text{min}$$

2. $h(x) = \sqrt{2+x^4}$

$$h'(x) = \frac{1}{2}(2+x^4)^{-\frac{1}{2}} \cdot 4x^3 \quad x=0 \text{ (min)}$$

3. $F(x) = -\frac{3}{1+x^2} = 3(1+x^2)^{-1}$

$$F'(x) = 3(1+x^2)^{-2} \cdot 2x \quad x=0$$

$\therefore \text{min}$

4. $H(x) = 50 \cdot e^{-2x^2}$

$$f'(x) = 4xe^{-2x^2} \quad \therefore \text{stationary point}$$

Economics Application

- Profit function $\pi(Q)$

$$\pi(Q) = R(Q) - C(Q)$$

A firm produces

$$Q = 5\sqrt{L}$$

units of output when L units of labour are employed.

The firm sells its output at a price of **€72 per unit**, and the wage rate is **€30 per unit of labour**.

(a) Write the profit function $\pi(L)$.

(b) Find the value of L that maximizes profit.

(c) Using the first-order condition, implicitly differentiate with respect to the wage rate w to find how the optimal labour choice L^* changes when w changes.

$$(a) \pi(L) = 72(5\sqrt{L}) - 30L$$

$$(b) 360\sqrt{L} - 30L$$

$$\text{FOC: } 180(L)^{-\frac{1}{2}} - 30 = 0$$

$$\frac{1}{L^{\frac{1}{2}}} = \frac{30}{180}$$

$$L^{\frac{1}{2}} = 6$$

$$L = 36$$

$$(c) 72(5\sqrt{L}) - \omega L$$

$$360\sqrt{L} - \omega L$$

$$\frac{180}{\sqrt{L}} - \omega = 0$$

$$\therefore \frac{dL}{d\omega} = \frac{L^{3/2}}{90} \quad \text{sub } L=36 \quad \frac{dL}{d\omega} = -\frac{12}{5}$$

A firm produces

$$Q = f(L)$$

units of output when L units of labour are employed, where

$$f'(L) > 0, \quad f''(L) < 0.$$

The output price is $p > 0$, and the wage rate is $w > 0$.

(a) Write the firm's profit function.

(b) Derive the first-order condition satisfied by the profit-maximizing labour input L^* .

(c) Totally differentiate the first-order condition to derive an expression for

$$\frac{dL^*}{dw}.$$

(d) Based on the assumptions on f , determine whether $\frac{dL^*}{dw}$ is positive or negative.

$$(a) \pi(L) = pf(L) - \omega L$$

$$(b) \pi'(L) = pf'(L) - \omega = 0$$

$$(c) pf''(L) \cdot \frac{dL}{d\omega} - 1 = 0$$

$$\frac{dL}{d\omega} = \frac{1}{pf''(L)}$$

(d) $\therefore$ negative

A monopolist faces the inverse demand function

$$P(Q) = 90 - 3Q$$

and has cost function

$$C(Q) = 12Q + 20.$$

(a) Write the profit function $\pi(Q)$.

(b) Find the profit-maximizing output Q^* .

(c) Find the associated monopoly profit $\pi(Q^*)$.

(d) Suppose the marginal cost parameter changes from 12 to k , so that

$$C(Q) = kQ + 20.$$

Find $\frac{d\pi(Q^*)}{dk}$.

$$\begin{aligned}\text{(a)} \quad \pi(Q) &= 90Q - 3Q^2 - 12Q - 20 \\ &= 78Q - 3Q^2 - 20\end{aligned}$$

$$\begin{aligned}\text{(b)} \quad \pi'(Q) &= 78 - 6Q = 0 \\ 6Q &= 78 \\ Q &= 13\end{aligned}$$

$$\text{(c)} \quad \pi(Q) = 1014 - 507 - 20 = 487$$

$$\begin{aligned}\text{(d)} \quad \pi(Q) &= 90Q - 3Q^2 - kQ - 20 \\ \pi'(Q) &= 90 - 6Q - k = 0 \\ Q^* &= \frac{90-k}{6}\end{aligned}$$

$$\pi(Q) = (90-k)Q - 3Q^2 - 20$$

$$\begin{aligned}
 \pi(Q^*) &= (90-k)\left(\frac{90-k}{6}\right) - 3\left(\frac{90-k}{6}\right)^2 - 20 \\
 &= \frac{(90-k)^2}{6} - \frac{3(90-k)^2}{36} - 20 \\
 &= \frac{(90-k)^2}{6} - \frac{(90-k)^2}{12} - 20 \\
 &= \frac{(90-k)^2}{12} - 20
 \end{aligned}$$

$$\begin{aligned}
 \frac{d\pi'(Q^*)}{dk} &= \frac{-2(90-k)}{12} \\
 &= -\frac{1}{6}(90-k)
 \end{aligned}$$

A monopolist faces inverse demand

$$P(Q) = b - Q$$

where $b > 0$, and has cost function

$$C(Q) = kQ$$

with $0 < k < b$.

(a) Write down the profit function $\pi(Q)$.

(b) Find the profit-maximizing output Q^* .

(c) Find the monopoly profit evaluated at the optimum, $\pi(Q^*)$.

(d) Differentiate $\pi(Q^*)$ with respect to k .

(e) Interpret the sign of $\frac{d\pi(Q^*)}{dk}$.

$$(a) \pi(Q) = bQ - Q^2 - kQ$$

$$(b) \pi'(Q) = b - 2Q - k = 0$$

$$b - k = 2Q$$

$$Q^* = \frac{b-k}{2}$$

$$(c) \pi(Q^*) = (b-k)Q - Q^2$$

$$= (b-k) \left(\frac{b-k}{2} \right) - \left(\frac{b-k}{2} \right)^2$$

$$= \frac{(b-k)^2}{2} - \frac{(b-k)^2}{4}$$

$$= \frac{(b-k)^2}{4}$$

$$(d) \frac{d\pi(Q^*)}{dk} = \frac{-2(b-k)}{4}$$

$$= -\frac{b-k}{2}$$

$$(e) \quad b - k > 0$$

$$-(b - k) < 0$$

$$-\frac{(b - k)}{2} < 0$$

$$\frac{d\pi(Q^*)}{dk} < 0$$

As $k \uparrow$, $\pi(Q^*)$ decreases

- A monopolist faces inverse demand

$$P(Q) = a - bQ$$

- with $a > 0$, $b > 0$, and constant marginal cost k , where $0 < k < a$.

- **(a)** Write the profit function.

- **(b)** Find the profit-maximizing output Q^* .

- **(c)** Find the corresponding maximum profit $\pi(Q^*)$.

- **(d)** Compute

$$\frac{d\pi(Q^*)}{dk}.$$

$$(a) \pi(Q) = aQ - bQ^2 - kQ$$

$$(b) \pi'(Q) = a - 2bQ - k = 0$$

$$= a - k = 2bQ$$

$$Q^* = \frac{a-k}{2b}$$

$$(c) \pi(Q^*) = (a-k) \left(\frac{a-k}{2b} \right) - b \left(\frac{a-k}{2b} \right)^2$$

$$= \frac{(a-k)^2}{2b} - \frac{b(a-k)^2}{4b^2}$$

$$= \frac{(a-k)^2}{2b} - \frac{(a-k)^2}{4b}$$

$$= \frac{(a-k)^2}{4b}$$

$$(c) \frac{d\pi(Q^*)}{dk} = \frac{-2(a-k)}{4b} = -\frac{(a-k)}{2b}$$